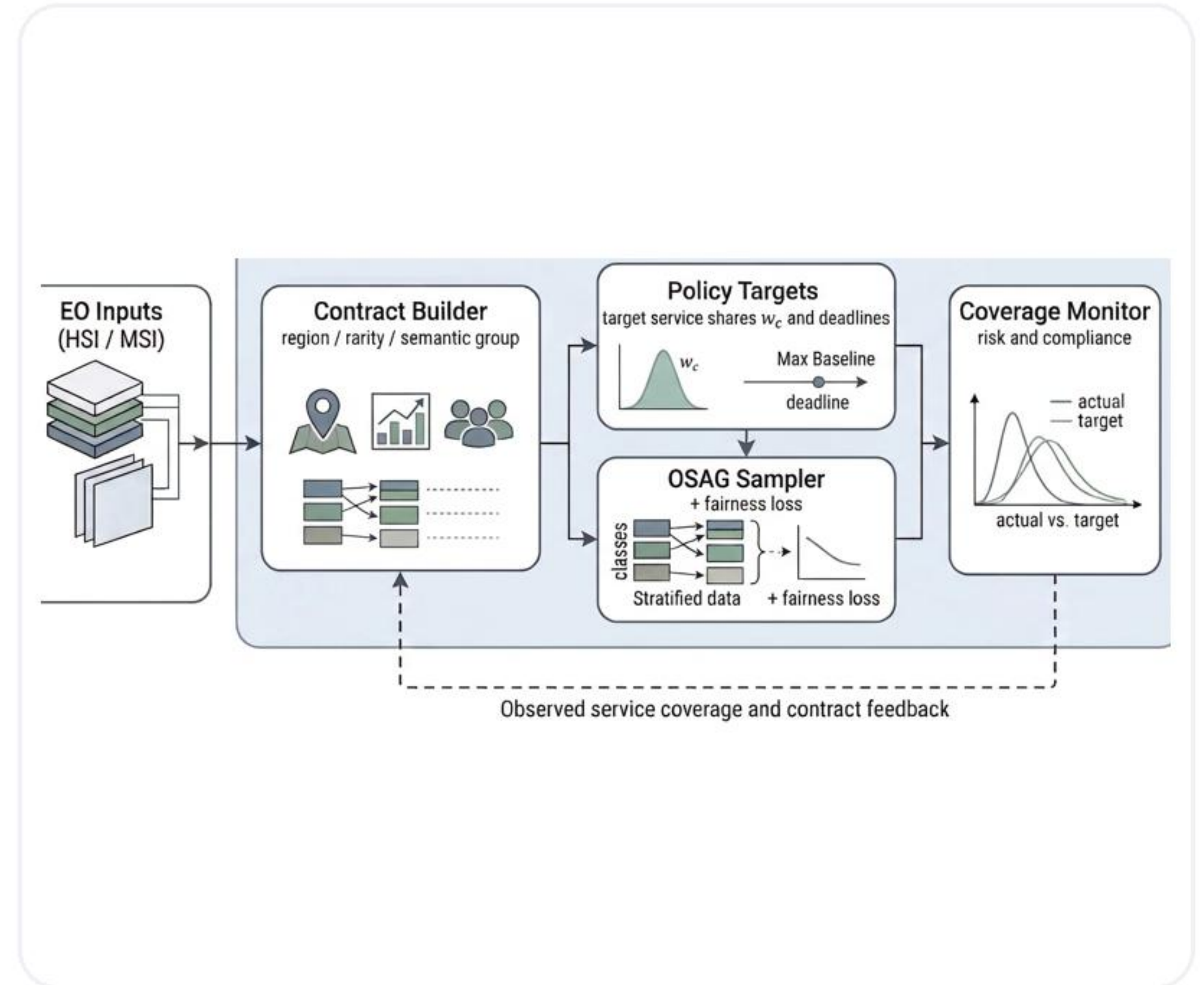


Governance-Aware Earth Observation Learning

Contract-Governed Training with Observed Service Agreement Graphs

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Background and Motivation

Why accuracy alone is not enough

- EO systems now support agriculture, land-use auditing, disaster response, and regional planning.
- In these settings, average accuracy matters, but service allocation also matters: which regions, classes, or semantic groups receive attention during training.
- Standard empirical risk minimization does not expose that allocation explicitly.
- A model can therefore look strong globally while still under-serving the contracts that matter most.

Problem in one sentence

The key question is not only whether the model performs well on average, but also who is being served during training.

What this thesis changes

Accuracy-only view

good global score

Can still hide weak service for small but mission-critical contracts.

Governance view

explicit service

Tracks whether observed exposure matches the intended contract policy.

The thesis adds a lightweight governance layer on top of ordinary EO training rather than replacing the classifier.

Research Gap and Questions

Central gap

Ordinary EO training optimizes loss over samples, but it does not represent service allocation explicitly or make policy compliance measurable.

RQ1

How can EO training be reformulated so that service allocation over semantically meaningful contracts becomes explicit and measurable?

RQ2

Can a lightweight governance layer drive empirical coverage toward policy-defined target shares without requiring a new backbone architecture?

RQ3

What is the empirical relationship between governance quality and predictive utility on HSI and MSI benchmarks?

RQ4

Can the thesis provide a reliable runnable demo that communicates the core idea clearly in a defense setting?

Thesis Contributions

Conceptual

Reframes EO training as a governance-aware learning problem in which training exposure becomes a measurable service object.

Technical

Develops OSAG as a contract-aware sampling and monitoring layer with target shares, alpha control, and λ_C fairness pressure.

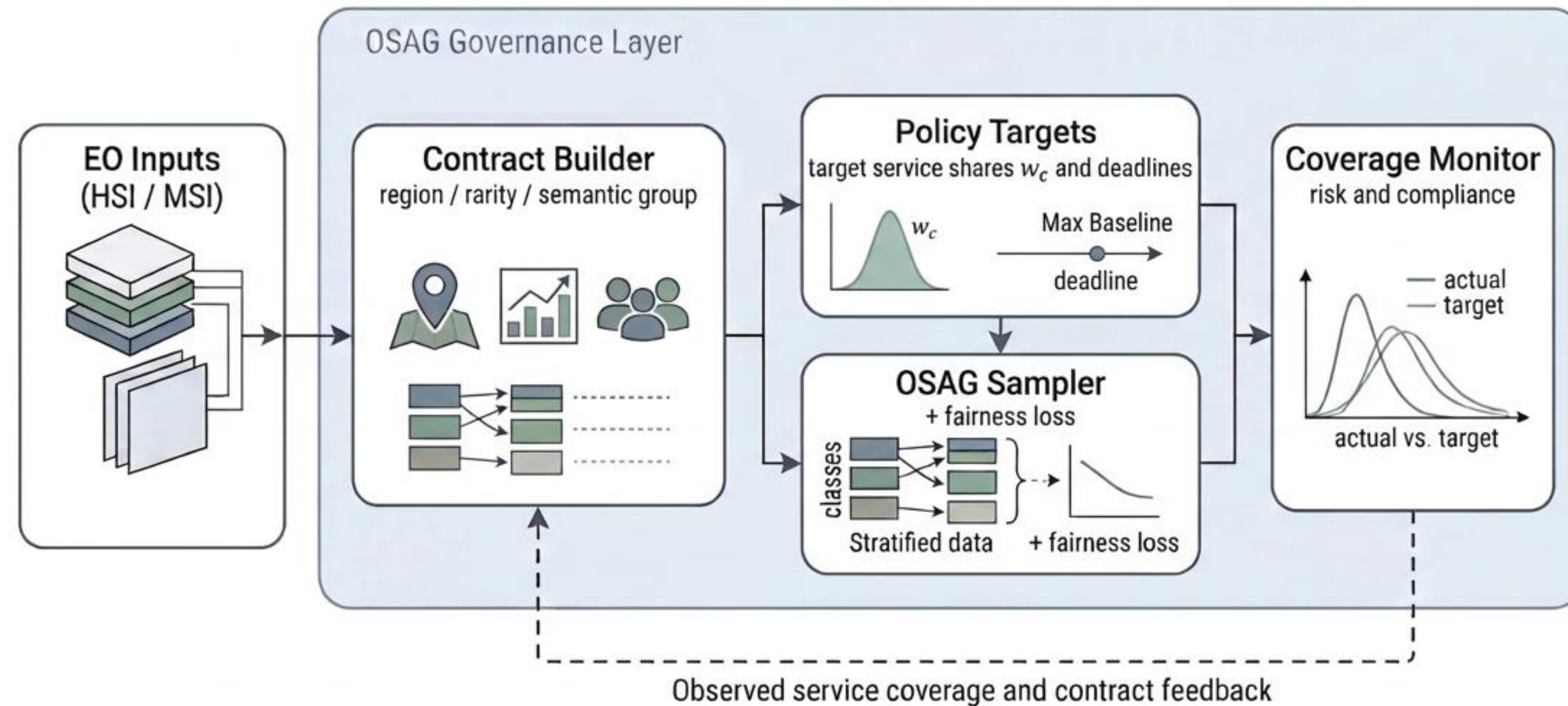
Empirical

Provides fresh rerun evidence on corrected Indian Pines, corrected Salinas, and canonical EuroSAT MSI, plus ablation, runtime, and a scoped ResMLP check.

Defense artifact

Adds a runnable dispatch demo and a reproducibility appendix so the thesis can be defended through both formal evidence and a clear operational story.

Figure 1.1 Method Overview



How to read the pipeline

- EO inputs are first mapped into semantically meaningful contracts.
- Policy targets assign an explicit target service share to each contract.
- The OSAG sampler injects that policy into training exposure without changing the backbone family.
- The coverage monitor reports whether observed exposure matches the intended allocation.
- The feedback loop turns service allocation into something visible, auditable, and controllable.

One-sentence takeaway

OSAG is a governance layer around training, not a replacement for the classifier itself.

Core Concepts

Contract

A service unit that is meaningful to the application, such as a spatial cell, a semantic group, or a rarity-aware group.

Target service share

The intended fraction of training attention assigned to each contract.

Empirical coverage

The observed fraction of training steps actually spent on each contract.

PrioCovErr

The L1 gap between empirical coverage and the target distribution. Smaller means better policy compliance.

alpha

The sampler-mixing control between baseline sampling and pure OSAG sampling.

lambda_C

A fairness penalty that adds pressure when high-priority contracts remain harder in loss space.

Benchmarks and Contract Design

Formal benchmark inputs

- HSI benchmark: corrected Indian Pines and corrected Salinas, aligned to a common band subset and merged into one joint setting.
- MSI benchmark: canonical 13-band EuroSAT MSI from dataset/EuroSAT_MS.zip.
- Main benchmark tables are fresh five-seed reruns from the formal script pipeline.
- The main benchmark uses a lightweight two-hidden-layer MLP so the governance effect remains interpretable.

Realized contract schemas

Case	Schema	Count
HSI	dataset + 4x4 grid cell + rare-class flag	44
EuroSAT coarse	four semantic service groups	4
EuroSAT fine	semantic group + training-split rare flag	6

Target-share policy

Target service share is set by $w_c = (p_c n_c) / \sum(p_{c'} n_{c'})$.

- HSI: rare contracts use priority 3; others use priority 1.
- EuroSAT coarse: urban and water use priority 2; agri and natural use priority 1.
- EuroSAT fine: rare contracts use priority 2; others use priority 1.

Main Benchmark Results

All values are reported in percent, except PrioCovErr, which is shown in percentage points to match the thesis tables.

Indian+Salinas fresh five-seed summary

Policy	Acc_all	Acc_high	PrioCovErr
Random	91.47 +/- 0.17	82.97 +/- 4.73	23.49 +/- 0.02
Class-Balanced	90.25 +/- 1.71	86.26 +/- 3.80	31.37 +/- 0.05
OSAG-Mix (a=0.50)	91.68 +/- 0.25	86.51 +/- 4.46	11.75 +/- 0.06
OSAG	90.85 +/- 0.42	88.83 +/- 4.70	0.26 +/- 0.06
OSAG-FairLoss (l=0.5)	91.10 +/- 0.44	91.15 +/- 6.75	0.26 +/- 0.06

EuroSAT MSI fresh five-seed summary

Policy	Acc_all	Acc_high	PrioCovErr
Random	86.93 +/- 0.31	73.68 +/- 1.70	23.81 +/- 0.08
Class-Balanced	85.91 +/- 0.91	74.25 +/- 2.08	17.18 +/- 0.10
OSAG-Mix (a=0.75)	86.70 +/- 0.53	74.75 +/- 1.18	5.96 +/- 0.12
OSAG	86.39 +/- 0.56	76.05 +/- 1.80	0.20 +/- 0.08
OSAG-FairLoss (l=0.5)	86.30 +/- 0.52	77.04 +/- 2.65	0.20 +/- 0.08

Policy-alignment gain

about 24 -> 0.2-0.3

On both benchmarks, OSAG-family operating points move policy misalignment close to zero.

Utility stays useful

Acc_all remains competitive

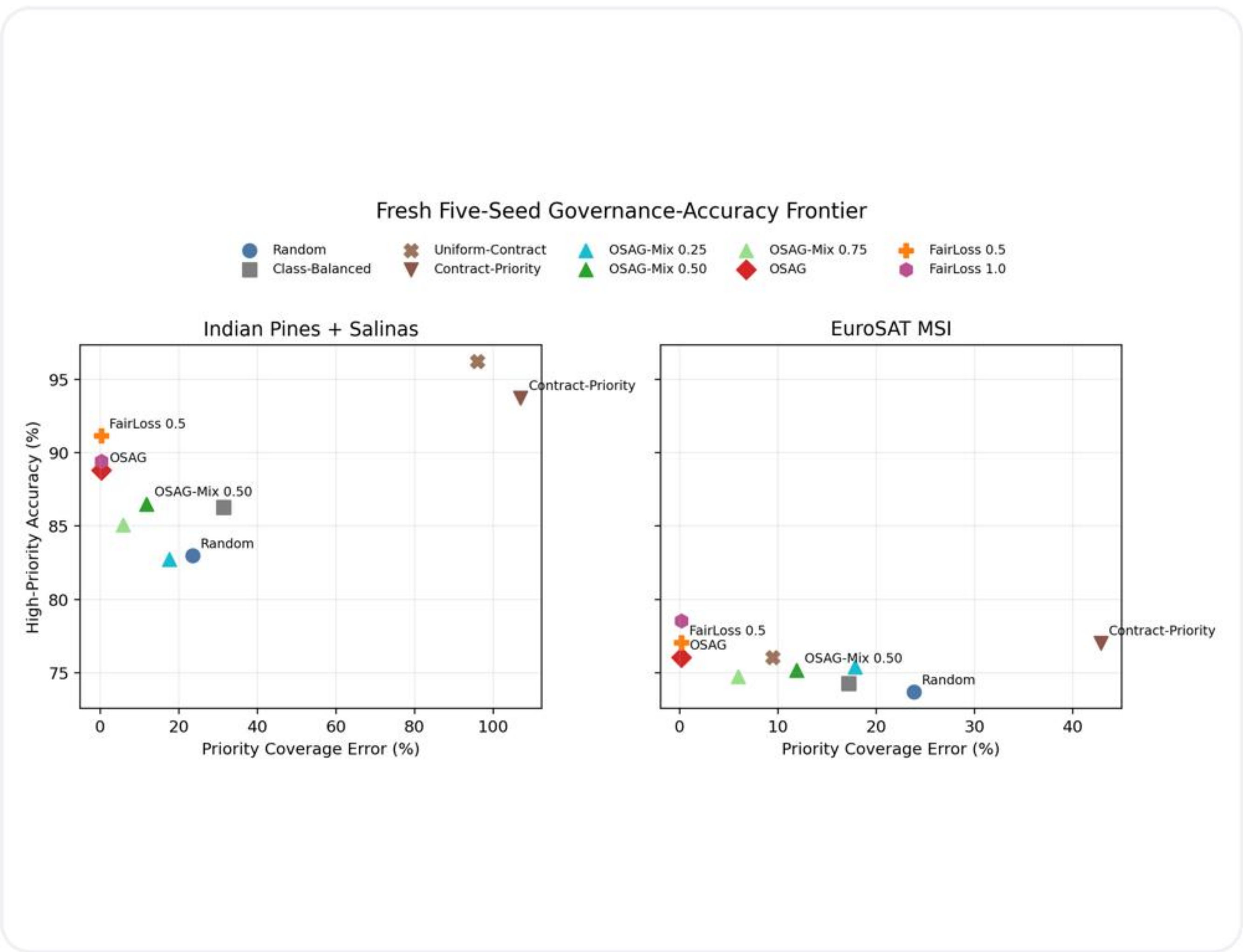
The governance gain is not bought by abandoning overall accuracy.

High-priority utility

Acc_high improves

OSAG-FairLoss gives the strongest highlighted Acc_high operating points.

Why Accuracy Alone Is Not Enough



Read Table 5.2 correctly

The reported deltas are absolute percentage-point differences relative to Random, not relative percentage changes.

Direct OSAG vs Random examples

Dataset	Delta Acc_all	Delta Acc_high	Delta PrioCovErr
Indian+Salinas	-0.62	+5.85	-23.23
EuroSAT MSI	-0.54	+2.36	-23.61

Interpretation

The thesis contribution is governance-aware training: OSAG reaches operating points that are much closer to the intended contract policy while preserving useful predictive utility.

Contract Design, Runtime, and Robustness

EuroSAT contract-design ablation

Fresh three-seed rerun. Fine contracts preserve more accuracy while still reaching near-zero policy error.

Variant	Rnd Acc_all	OSAG Acc_all	Rnd PrioCovErr	OSAG PrioCovErr
Coarse	86.98	86.03	33.29	0.12
Fine	86.98	86.28	23.80	0.21

- Fine contracts start from a lower Random PrioCovErr baseline.
- The accuracy cost of governance is smaller in the fine design than in the coarse design.
- Contract design therefore changes the cost of policy alignment.

Runtime interpretation

EuroSAT per epoch

Random 1.367 s | OSAG 1.358 s

The practical overhead is comparable on the MSI benchmark.

HSI per epoch

Random 3.524 s | OSAG 2.938 s

Random is slower in this rerun mainly because of one outlier seed, not because OSAG is intrinsically faster.

Runtime reading

The thesis treats runtime as comparable-overhead evidence. It does not claim that OSAG is a speed optimization.

Scoped ResMLP robustness check

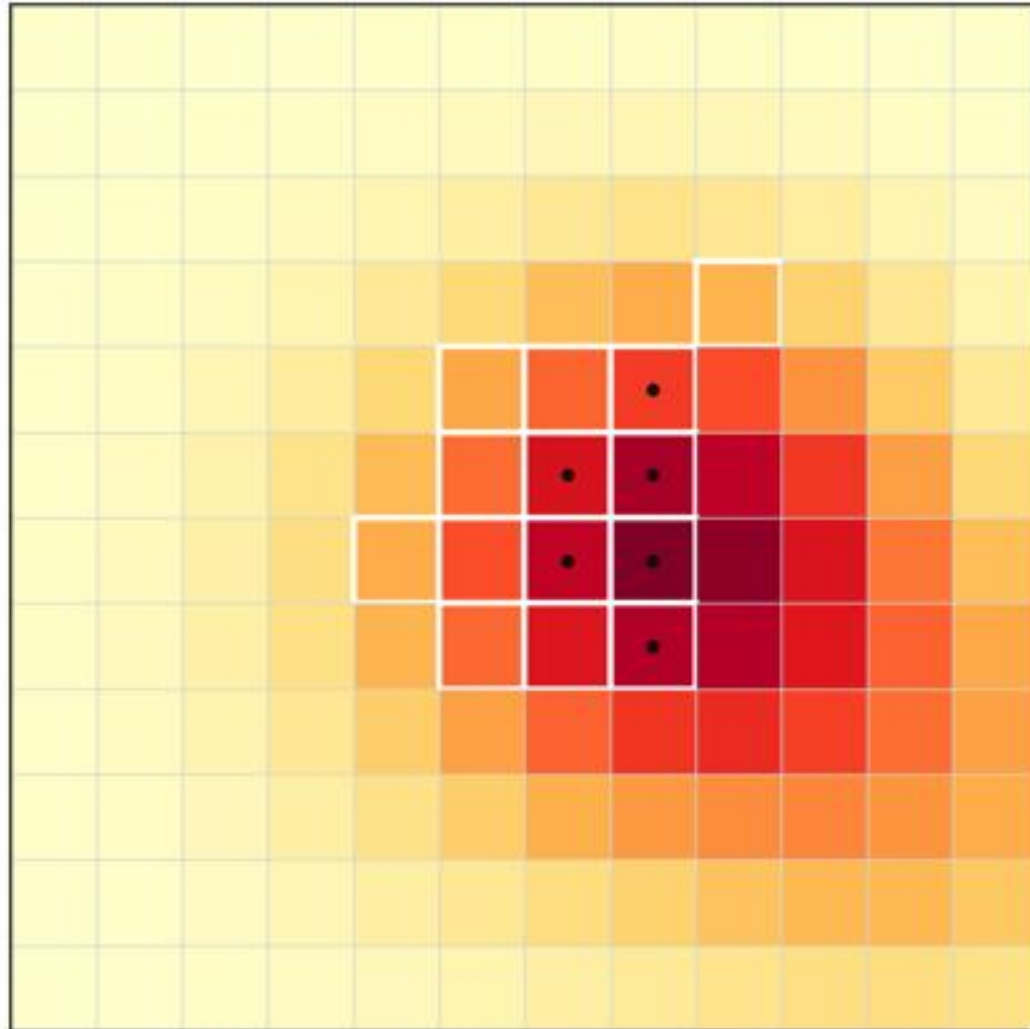
EuroSAT only. ResMLP is a robustness extension, not a replacement for the main benchmark design.

Policy	Backbone	Acc_all	Acc_high	PrioCovErr
Random	Simple	86.93	73.68	23.81
Random	ResMLP	86.34	73.51	23.79
OSAG	Simple	86.39	76.05	0.20
OSAG	ResMLP	86.26	74.10	0.15

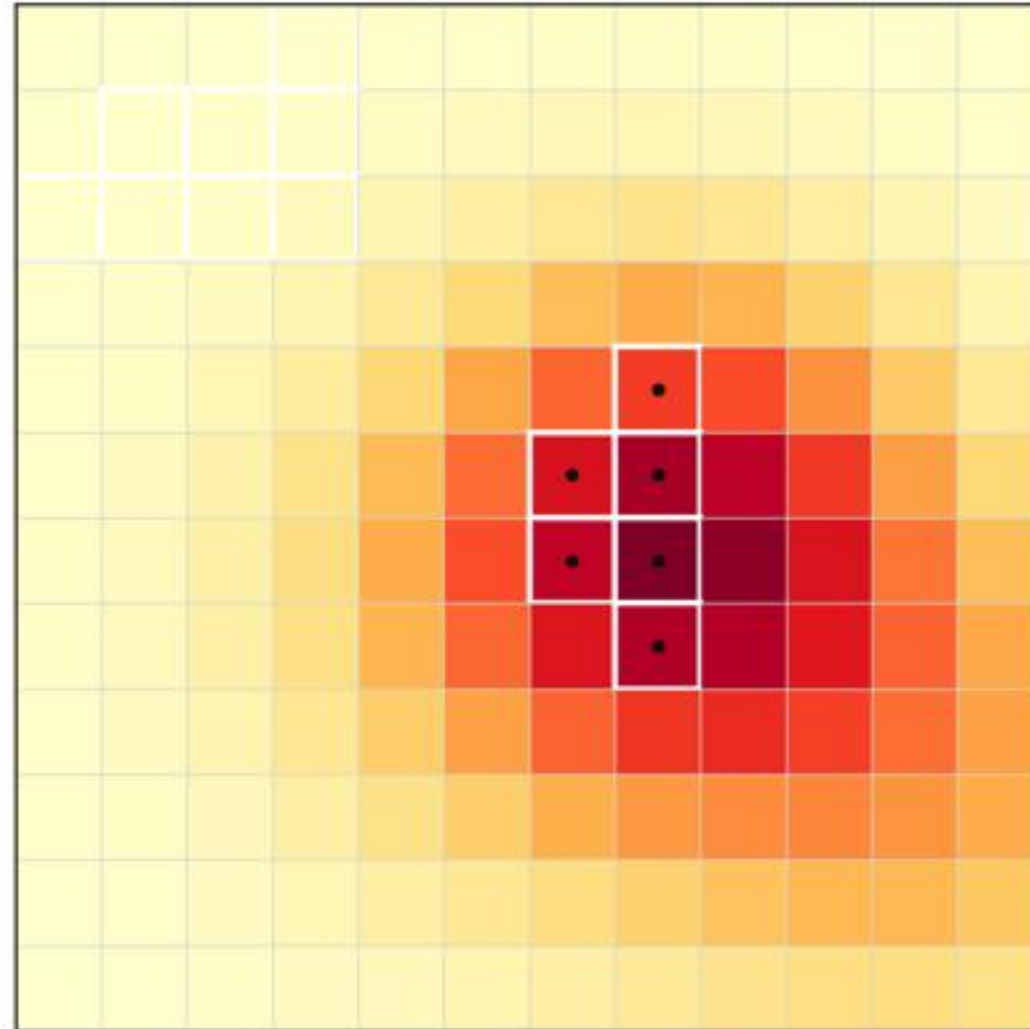
- Absolute utility shifts slightly with the stronger backbone.
- The core governance conclusion survives: OSAG still keeps PrioCovErr near zero.
- The thesis is not claiming state-of-the-art backbone performance.

Why the Demo Matters

Step 8: Contract-Priority



Step 8: OSAG



Different role from the benchmark

- The benchmark chapter provides the formal scientific evidence.
- The dispatch demo shows how the same governance idea behaves under budgets, deadlines, missed service, and time pressure.
- It makes contract-level behavior visible in a way that static tables cannot.

What the demo adds

- A 12 x 12 emergency grid over ten decision steps.
- Observation budget = 14 tiles and annotation budget = 6 tiles per step.
- Six contracts with explicit priorities, target weights, and deadlines.
- Operational metrics for compliance, coverage error, missed service, Q_{high} , and reliability.

Engineered Demo View

Why this support package helps

The engineered demo turns the same governance logic into a guided presentation view: primary KPIs on top, side-by-side maps in the middle, and one policy-gap chart below.

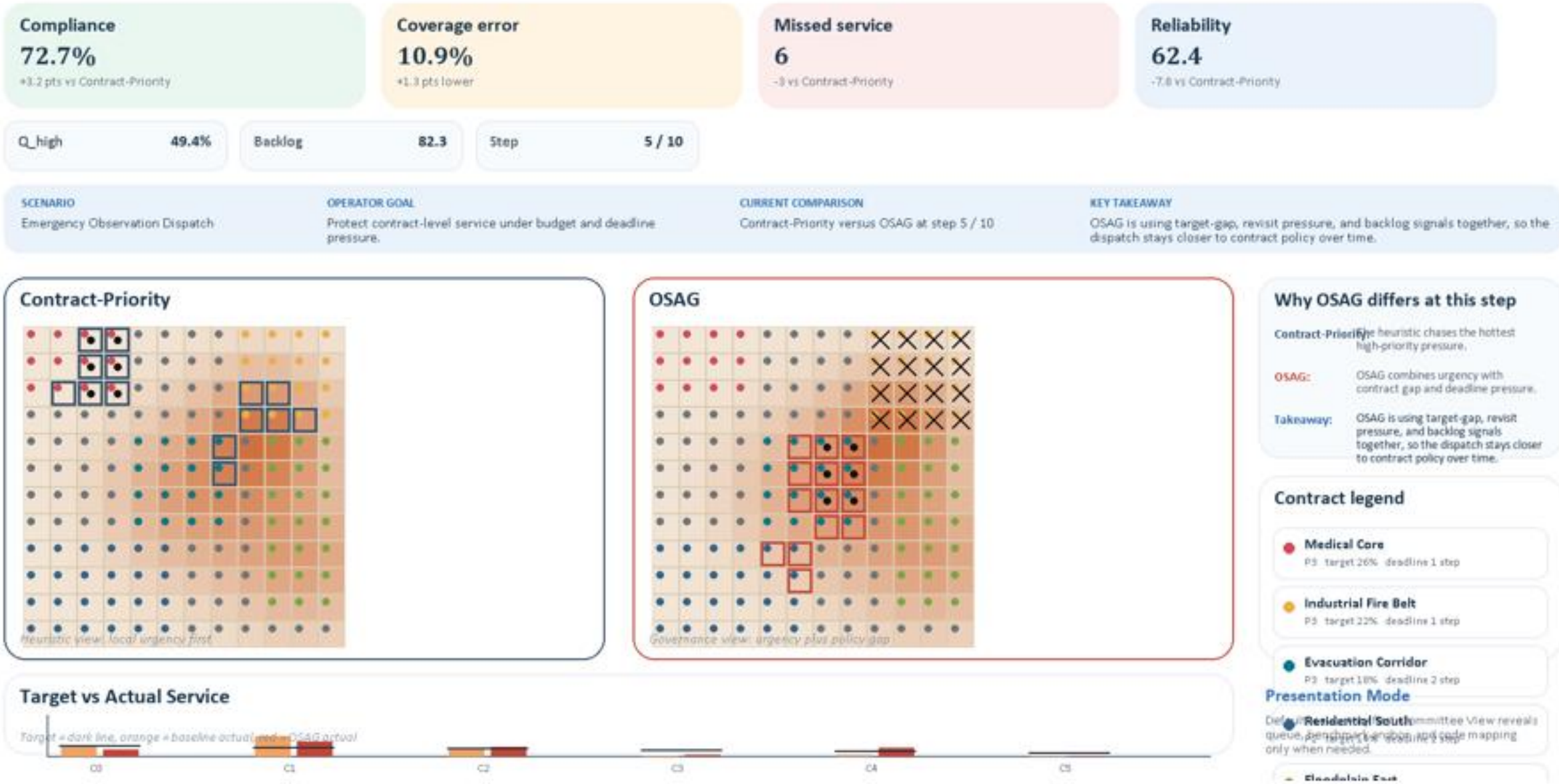
- Presentation mode emphasizes only the primary KPIs: compliance, coverage error, missed service, and reliability.
- Committee view can still expose queue details, evidence anchor, and code mapping without leaving the page.
- The package is fully static HTML/CSS/JS: no backend, no training, and no network are required on defense day.
- The formal benchmark remains the evidence layer; the engineered demo is the explanation layer.

ENGINEERED DEMO V2

Emergency Observation Dispatch Overview

Protect mission-critical contracts under fire, flood, and evacuation pressure.

Maintain contract-level service while sudden hazards and one-step deadlines compete for a fixed observation and annotation budget.



Engineered Demo V2 emergency scenario overview used as defense-day support material.

Code Architecture and Implementation Path

Formal benchmark path

This is the main evidence chain that produces the fresh results, tables, and figures.

`scripts/run_all_real_experiments.py`

`scripts/reproduce_osag_real.py`

`build_contract_table_from_meta(...) / run_one(...) / compute_coverage_errors(...)`

`results/raw/, results/tables/, results/runtime/, results/figures/`

Dispatch demo path

This is the live operational explanation path used for the dashboard and stepwise behavior.

`osag_demo/run_visual_demo.py`

`osag_demo/run_dispatch_demo.py`

`osag_demo/dispatch_demo_core.py / choose_observations(...)`

`osag_demo/dispatch_demo_assets.py -> demo_dashboard.html`

Defense asset path

This branch turns saved outputs and thesis assets into defense-ready materials.

`LaTeX thesis figures and tables`

`demo support package assets`

`current short deck structure reference`

`final clean deck, notes, audit, and rebuild report`

Safe live-inspection order

If the committee asks to see code, start from the benchmark entry, then the contract and coverage functions, then the demo entry, and finally `choose_observations(...)` in the dispatch core.

Limitations and Claim Boundaries

What the thesis supports

- OSAG can strongly reduce contract-level policy misalignment on the current HSI and EuroSAT MSI benchmarks.
- The governance effect survives fresh reruns, contract ablation, runtime scrutiny, a scoped ResMLP extension, and the dispatch demo.
- The evidence chain is script-driven and traceable through saved outputs, manifests, logs, configs, and thesis-ready assets.

What stays explicitly bounded

- The graph is a modeling and control view over the contract space, not a full explicit graph optimizer in the current implementation.
- The HSI split is pixel-stratified rather than spatially blocked, so absolute accuracy may be optimistic relative to stricter deployment conditions.
- The main benchmark uses a lightweight MLP to isolate governance effects; this is not a backbone state-of-the-art study.
- The reproducibility boundary is artifact-driven and script-driven; the thesis workspace itself is documented through manifests and reports rather than an internal git ledger.
- The demo is a supportive operational artifact, not the main benchmark evidence.

Reproducibility and Public Release

1. Fully rerun main benchmark evidence

Indian Pines + Salinas and EuroSAT MSI main results are freshly rerun over five seeds from formal scripts and canonical local inputs.

2. Fully rerun extension evidence

The EuroSAT coarse-versus-fine ablation is freshly rerun over three seeds, and the EuroSAT ResMLP extension is freshly rerun over five seeds.

3. Artifact-level reproducibility

The dispatch demo, thesis figures and tables, and thesis PDF can all be regenerated locally.

Formal inputs, outputs, and public companion release

- Formal inputs live under dataset/: Indian_pines_corrected.mat, Indian_pines_gt.mat, Salinas_corrected.mat, Salinas_gt.mat, and EuroSAT_MS.zip.
- Fresh outputs are written under results/, logs/, artifacts/, and osag_demo/visual_outputs/.
- Companion public source release: github.com/dqswordman/osag-eo-governance

Final Takeaways and Q&A

1. EO training is also a service-allocation problem.

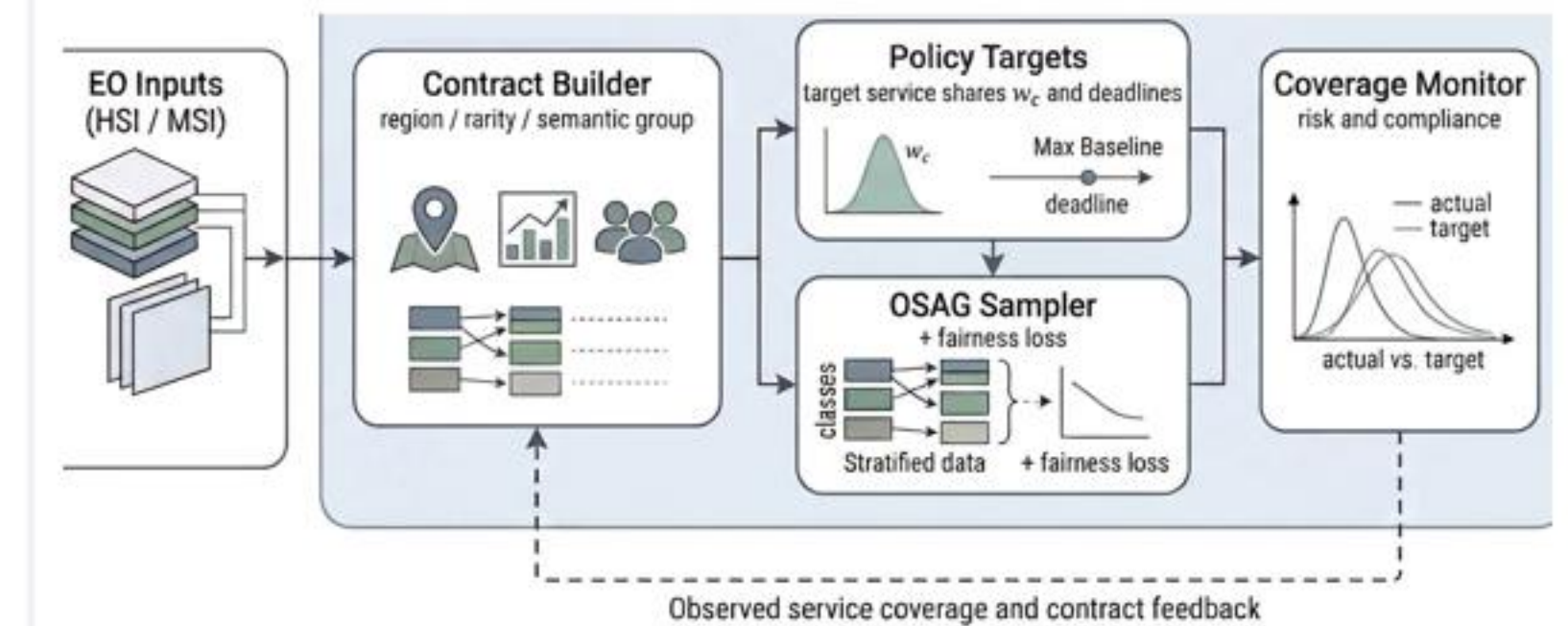
The thesis makes that allocation explicit through contracts and target service shares.

2. OSAG sharply reduces policy misalignment.

On the current HSI and EuroSAT MSI benchmarks, PrioCovErr falls from about 24 points to about 0.2-0.3 points.

3. The defense story is complete and bounded.

Fresh reruns, demo support, implementation path, limitations, and Appendix A together make the thesis inspectable and defensible.



Questions are welcome

I am ready to answer questions about the method, the results, the demo, the code path, or the claim boundaries.